

Science 24

UNIT C WORKBOOK



SENIOR HIGH SCHOOL

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Chapter 9: Disease

Define the following:

Pasteurization

Disease

Microbes

Pathogens

Bacteria

Viruses

Fungi

Protist

Parasites

Communicable Disease

Non-communicable Disease

Botulism

Salmonellosis

Sanitary

Poverty

Staph bacteria

Sterilization

Aseptic

1. Describe 5 factors that may interact with one another and possibly make you sick.
(pg. 158)

2. What did the following scientists discover.

a. Robert Koch (pg. 158)

b. Louis Pasteur (pg. 158)

c. Florence Nightingale (pg. 170)

d. Joseph Lister (pg. 170)

3. List the conditions that bacteria need to grow. Explain why each is important. (pg. 159)

4. What is a pathogen? Give three examples of pathogens. (pgs. 159 to 162)

5. Explain how viruses such as the AIDS virus spread. (pg. 159)

6. How do protists spread? (pg. 162)

7. What causes beaver fever? (pg. 162)

8. What causes athletes foot? (pg. 162)

9. What might happen to the bacteria in a hamburger you left on the counter for a few hours? Explain why. (pg. 159)

10. List 3 types of communicable disease. (pg. 163)

11. List 3 types of non-communicable disease. (pg. 163)

12. How does a clean environment protect you from disease? (pg. 164)

13. List four factors that affect the risk of developing a non-communicable disease. (pg.165)

14. What is poverty and how does it contribute to the spread of disease? (pg. 166)

15. What types of disease kills most people in poor countries? In rich countries?
Explain why you think this is so. (pg. 167)

16. Explain how each of the following methods of food preservation protects food from pathogens: drying, canning, salting, smoking, refrigeration, freezing, chemical additives, vacuum packing, irradiation, pasteurization. (pg. 168)

17. List 5 recommendations for protecting food preparation surfaces and utensils from contamination. (pg. 169)

18. Explain why you would put an antiseptic and a disinfectant in a first aid kit. (pg.170)

19. What is the purpose of sterilization? (pg. 170)

20. List 4 aseptic practices. (pg. 170)

Chapter 10: Society's Response to Disease

Define the following:

Pandemics

Epidemics

Quarantine

Public Health

Vaccinations

Migration (pg. 184)

AIDS (pg. 179)

Tuberculosis (pg. 178)

Bubonic Plague (pg. 178)

Advisory (pg. 185)

1. Who is responsible for the preserving the quality of water you use and drink? (pg. 174)
2. Why was smallpox tuberculosis and the flu so deadly to First Nations Peoples? (pg. 176)
3. What were the effects of disease on Aboriginal Groups? (pg. 176)
4. How is a pandemic different from other diseases? Give an example of a pandemic disease. (pg. 176)
5. When does an epidemic become a pandemic? (pg. 178)

6. Complete the following table:

	Bubonic Plague	Tuberculosis	Spanish Flu	AIDS
Social Conditions				
Impact of the pandemic				
When and where the disease was present				

7. Why did society begin to take action to protect the health of the community? (pg. 180)

8. What jobs are the provincial health department responsible for? (pg. 181)

9. What does the Regional Health Authority do when it comes to water quality? (pg. 181)

10. How do landfills prevent toxic chemicals from entering the water system? (pg. 181)

11. Why do we treat human and animal waste? (pg. 182)

12. If the air quality is between 50 to 100 (Poor to Very Poor) what kind of alert or warning do you think would be sent out? (pg. 182)

13. List 6 vaccinations that are given by the Health Authority. (pg. 182)

14. List the 4 points put forward by the BAC campaign. (pg. 183)

15. Complete the following table (pg. 184):

<u>Factor</u>	<u>Concern</u>	<u>Ways to reduce risk</u>
Population growth and overcrowding		• disease monitoring • Emergency response plans to deal with a community with disease
Migration	Diseases may be carried from country to country or from rural to urban	
	People eat out in restaurants	• Restaurant inspections • sanitary workplaces
Travel		
Food supply		• Food Inspectors check food supply • Transportation vehicles are refrigerated
Sexual activity and childbirth		

16. Do you think it is important to have public health guidelines? Use examples to help you explain why or why not. (pg. 187)

17. Read the Walkerton's Water Tragedy Timeline on page 186.

Do you think that the PUC manager should face criminal charges?

Explain why you think the way you do.

If you think that he should be charged what do you think he should be charged with?

Chapter 11: Protecting Yourself from Disease

Define the following:

Antibodies

Physical defenses

Inflammatory response

Immune System

Macrophages

Antigens

Inherited Immunity

Acquired Immunity

Active Immunity

Passive Immunity

Vaccine

Immunization

Over-the-Counter Drugs

Antibiotic

1. Where does penicillin come from? (pg. 192)

2. What is the first line of defense against infection? (pg. 194)

3. Complete the chart on Physical Defenses (pg. 194):

<u>Structure</u>	<u>Function</u>
	Contains hair and mucous, which trap pathogens before they can enter the respiratory system. Warm incoming air.
	Catch pathogens before they enter the body. May become infected to help the immune system resist and fight future infections. Store white blood cells.
Eyelids	
Tears	
	Protect the eye.
Hair	
	Contains wax which traps pathogens and particles.
	Consists of a thick barrier with an acidic surface that prevents pathogens from entering. Prevents loss of body fluids.
Stomach	

4. Why do smokers have a nagging cough? (pg. 194)

5. What is the 2nd line of defense? (pg. 195)
6. List 3 different things that can trigger an inflammatory response. (pg. 195)
7. What is happening when your skin swells, turns red, and gets warm around a sliver? (pg. 195)
8. What do macrophages destroy? (pg. 195)
9. What is the 3rd line of defense? (pg. 195)
10. Read the following scenarios and write whether the following have antigens. (pg. 197)
- a. A virus enters the blood stream.
 - b. A red blood cell travels from the heart to the lungs
 - c. Bacteria transferred from a piece of broken glass enters the body.
 - d. Cancer cells in the body replicate.
 - e. A macrophage consumes a damaged cell.

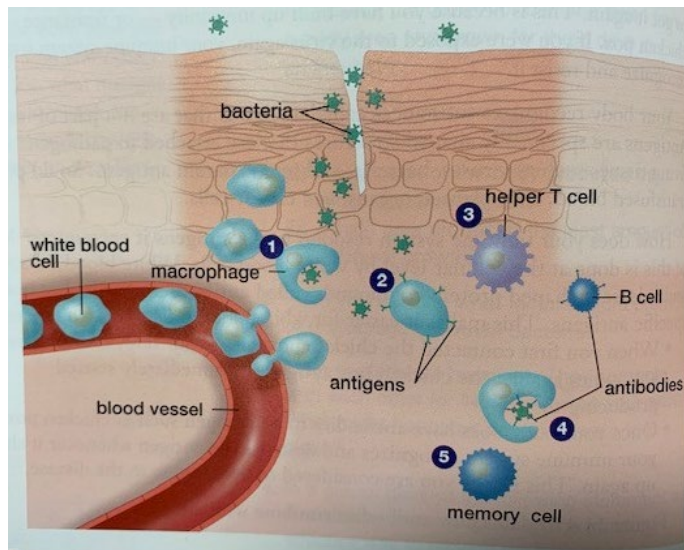
11. What does your body release when it senses an antigen? (pg. 197)

12. How do antibodies work to give us immunity? (pg. 197)

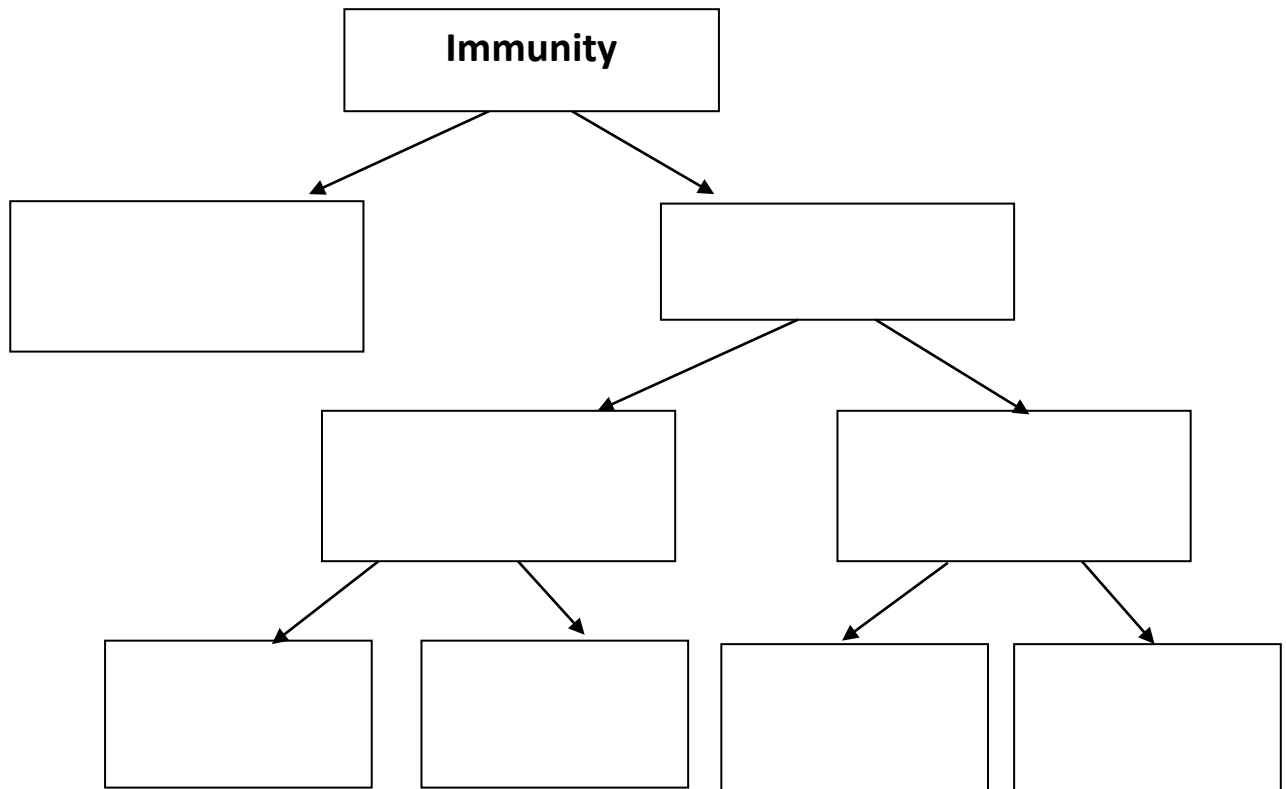
13. Look at the following diagram.

Order the following in proper chronological order from 1 to 5. (pg. 198)

- a. Antigens from dead bacteria stick out of the macrophage. _____
- b. Antibodies produced by B-cells attach to antigens and killer T-cells destroy them. _____
- c. Bacteria enter the body and macrophages surround them. _____
- d. B cells produce memory cells to store the information about how to kill the antigen so they can make the antibodies faster the next time. _____
- e. Helper T-cells read antigens and B-cells make and release antibodies. _____



14. Complete the following flowchart. (pg. 199)



15. Classify the following as a type of immunity. (pg. 199)

- A baby gets immunity from diseases from a mother's milk.
- You are vaccinated against smallpox.
- A cow is injected with snake venom and the antitoxin is given to you.
- You get chicken-pox and you never get it again.

16. What is the difference between passive and active immunity?
Give an example of each. (pg. 199)

17. Why do people with A blood reject transplants from people with B blood? (pg. 200)

18. Why are people with O negative blood considered universal donors? (pg. 200)

19. What is the difference between a vaccine and an antitoxin? (pg. 201)

20. How do vaccines work? (pg. 201)

21. Why should you get immunizations? Do you think some countries are justified in not giving them? (pg. 202)

22. How is your immune system related to allergies? (pg. 202)

23. What is the difference between over-the-counter drugs and antibiotics? (pg. 204)

24. Why do antibiotics not work on colds? (pg. 204)

25. List 3 problems with antibiotics. (pg. 205)

26. List 3 things you should do to use antibiotic safely? (pg. 206)

27. What should you do when choosing Over-The-Counter Drugs? (pg. 206)

Chapter 12: Genetics and Health

Define the following:

Genetics

Chromosomes

DNA

Genes

Heredity

Dominant

Hybrid

Purebred

Recessive

Genotype

Punnett Square

Pedigree

Mutation

Mutagens

Genetic Disorders

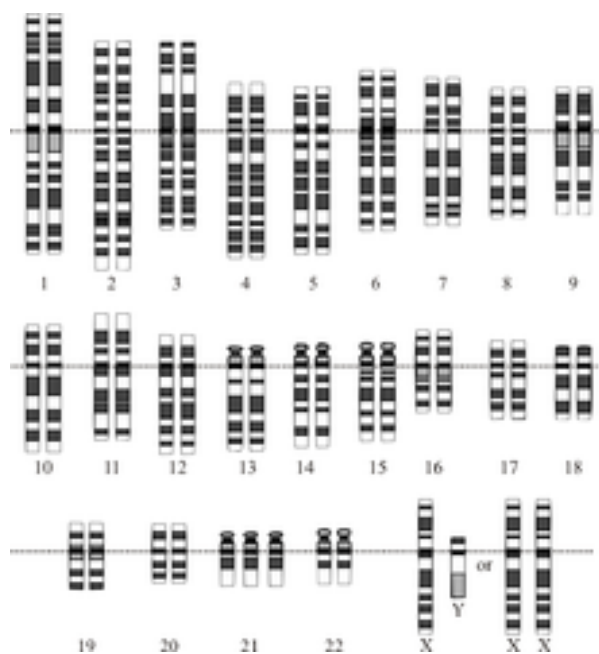
Genetic Engineering

Amniocentesis

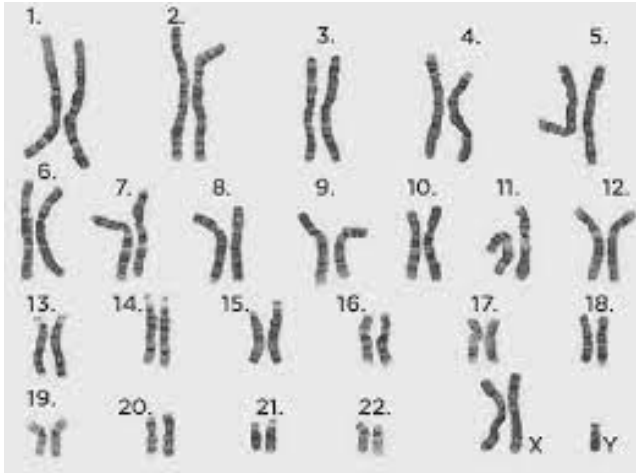
1. Where in the cell are the chromosomes? (pg. 212)
2. If one side of the DNA molecule looks like this A-G-C-T-G. What must the other side look like? (pg. 212)
3. How many chromosomes in a normal human? (pg. 213)
4. Here are 3 genetic disorders Identify what the person has using the following karyotypes.
Downsyndrome = Trisomy 21 (3 chromosomes), Turners Syndrome = a single X,
Kleinfelters Syndrome = XXY. (pg. 213)



Disorder:



Disorder:



Disorder:

5. For the following state whether the person is male or female. (pg.213)

A)XX

B)XY

6. Why can it be said that the male determines the sex of the baby? (pg. 213)

7. Who studied pea plants and became known as the father of genetics? (pg. 216)

Use the following Information to answer the next two questions.

TT or tt = Homozygous (two of the SAME genes)

Tt = Heterozygous (two different genes)

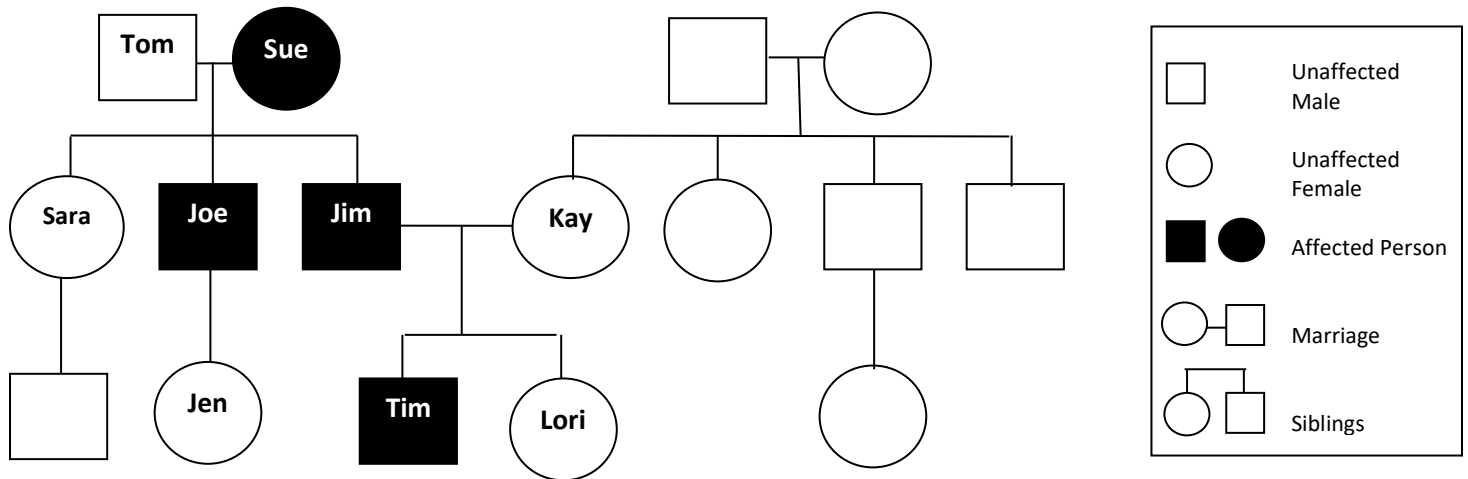
8. Make a Punnett Square using the following data. Tall Plants are Dominant to Short Plants. A heterozygous tall plant mates with a short plant. (pg. 217)
9. Brown Eyes are Dominant to Blue Eyes. If a father who is homozygous for Brown Eyes mates with a woman with Blue Eyes what chance does the offspring have of having blue eyes? (pg. 217)

10. Draw the punnett square for the following situation and fill out the chart below.
In sheep black wool is recessive to white. So two white heterozygous sheep mate.
(pg. 217)

	Genotype (The Genes)	Phenotype (What it looks like)	Percentage
Offspring 1			
Offspring 2			
Offspring 3			
Offspring 4			

What is the ratio of White sheep to Black Sheep?

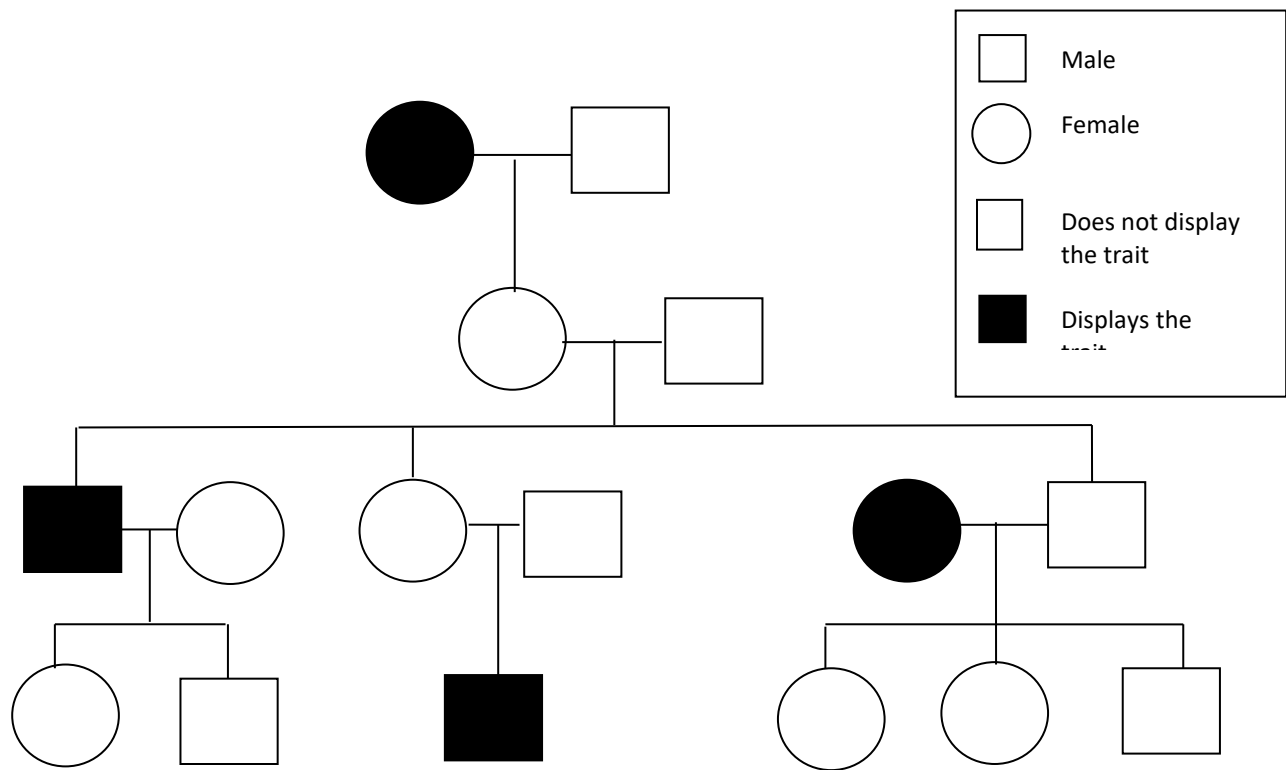
11. Look at the following pedigree. And answer the following questions. Pg 218



a) How many generations does this pedigree go?

b) Name the infected people.

c) How many children does Jim have?



12. Using the pedigree above determine if the infected trait is dominant or recessive.
EXPLAIN why. (pg. 218)

13. Construct a pedigree using the following information.
- a. Brown eyes is dominant to blue eyes
 - b. Mr. Pederson is homozygous for brown eyes marries a Blue-eyed Girl
 - c. They have 3 children, 2 sons and a daughter. They all have brown eyes.
 - d. One of the sons and the daughter do not have children.
 - e. 1 son marries a brown eyed girl. They have 2 sons and daughter.
Both sons have blue eyes.
 - f. Be sure to include a key. (pg. 218)

KEY:	
	
	
	
	

14. Describe how a mutation might occur when chromosomes are copied. (pg. 220)

15. a) What are mutagens? (pg. 220)

b) Name one example of a mutagen and a genetic disorder that results from it. Pg 222

16. Why should pregnant women not drink or do drugs? Pg 222

17. Why might amniocentesis pose an ethical dilemma for some people? (pg. 223)

18. Complete the following chart on advances in genetic research. (pgs. 223 and 225)

Pros of Genetic Research	Cons of Genetic Research

19. Should they allow cloning of people so people may harvest organs from their clone in the case of a medical emergency?